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Transmission line fault-cause classification based on multi-view sparse feature selection

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Abstract

With the development of communication and monitoring techniques, multi-source information in form of multi-view data is expected to support fault-cause identification for transmission line maintenance and fault disposal. However, the combined use of multi-view data poses a great challenge of information fusion. In response to this challenge, this paper proposes a multi-view sparse feature selection (MVSFS) method to combine contextual and waveform features for fault-cause classification. $l_{2,1}$ -norm sparsity regularization is adopted to select discriminative features across two views and an ε -dragging technique is integrated into the regression model to enhance its discriminant ability. Subsequently, an iteration optimization algorithm is devised to solve the model. Experiments on a real-life fault dataset demonstrate that contextual and waveform features can achieve higher accuracy than single view learning only through appropriate fusion, and the proposed MVSFS can overcome the drawback of conventional feature selection methods and make a great improvement of classification performance with multi-view data.

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1. Introduction

The increasing length and complexity of transmission networks coupled with their harsh operating environment are susceptible to various fault events that lead to disturbances or interruptions. In order to tackle fault events for line maintenance and reliable power supply, it is necessary to classify root causes of faults by their signatures. Transmission line faults may be caused by natural phenomena, creature activities and equipment failures, which are usually identified through expert analysis based on weather and fault behavior. However, manual diagnosis based on expert knowledge is time-consuming and places a great demand on personnel quality. With the increasing monitoring techniques and advanced recording equipment being implemented to support fault diagnosis, associated multi-source fault data are too voluminous to interpret manually.

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Several studies have been carried out to achieve line fault cause classification using data-driven methods. Rule-based methods, such as CN2 rule induction [1] and fuzzy logic [2,3], have been applied to identify fault causes with the intervention of domain experts. Due to the limitations of the fault scene and thresholds, rule-base methods have limited scalability and robustness. Machine learning (ML) methods are expected to classify fault causes automatically. In [4], the authors combine logistic regression (LR) and artificial neural network (ANN) to classify animal-caused and tree-caused outages with contextual features. In [5], a cause identifier based on deep belief network (DBN) is developed and original waveform characteristics are extracted from current traveling wave. The authors in [6] provide waveform signatures based on root mean square (RMS) and discrete wavelet transform (DWT) analysis and differentiate four types of cause using decision tree (DT). In [7], authors integrate historical outage, load and weather data to form a contextual feature set and make classification with ensemble tree models.

As shown in the mentioned literatures, contextual and waveform features are both able to realize a high degree of accuracy without each other. However, there are high data requirements such as ample field context data and high-resolution recording signal. In order to develop fault cause classification based on common data condition, we believe the combined used of contextual and waveform features can improve the accuracy and the challenge of information fusion need to be addressed. The authors in [8] first differentiate two feature types for comparison and reveal that feature concatenations even hamper the classification performance. Few studies make a further study on using both waveform and contextual characteristics for performance improvement.

To cope with this challenge, multi-view learning is introduced to fuse contextual and waveform features since they describe the same event from different perspectives. In this paper, a multi-view sparse feature selection (MVSFS) method is proposed for transmission line fault cause classification. Firstly, two-view feature sets are formed and composed of candidate features that respectively extracted from fault context and waveform. Then, the MVSFS method is applied to investigate the optimal feature combination. Specifically, ε -dragging is used to enlarge the margin between classes and a sparse regression model is developed based on $l_{2,1}$ -norm penalty terms. Finally, ML classifiers are employed for final classification with the integrated features. The capability and performance of the proposed MVSFS are verified by comparative experiments on real-life fault data.

2. Waveform and context characterization

This section provides the characterization of associated fault signals and contexts, and makes up the candidate feature pools for further combination. The first step of waveform characterization is acquiring the valid waveform segment from a series of fault signals. Nowadays, digital fault recorders are able to mark the triggering point for fault disturbances so we determine the beginning of a valid waveform segment accordingly. To avoid the extra disturbance caused by tripping, one and a half cycles after the tripping enabling signal is determined as the end of a segment.

This paper extends previous research work on waveform signatures and extracts the following waveform features for further selection [1,6,8–11]. Karenbauer transformation is applied to calculate three sequence components of three phase voltage/current. The waveform signals are denoted by W and sequence components are denoted by subscript s .

- (1) Maximum change of sequence components:

$$W_{s(\max)} = \frac{\max(|W_s - W_{s(\text{pre-fault})}|)}{W_{s(\text{pre-fault})}}, s = 0, 1, 2 \quad (1)$$

- (2) Maximum rate of change of sequence components:

$$\Delta W_{s(\max)} = \frac{\max(|\Delta W_s|)}{W_{s(\text{pre-fault})}}, s = 0, 1, 2 \quad (2)$$

- (3) Sequence component values at t -cycle: The values are measured at the point of t cycle from the start.

$$W_{s(t)} = \frac{W_{s(t)}}{W_{s(\text{pre-fault})}}, s = 0, 1, 2; t = 0, 0.5, 1, 1.5 \quad (3)$$

- (4) DC and harmonic content: DC and harmonic components are separated from the fundamental component via discrete wavelet transform. DC content is calculated as the ratio of the DC component to the fundamental component. Harmonic contents are the ratios of k th harmonic components to the fundamental component, $k \in [3, 5, 7, 9, 11]$.

- (5) Wavelet energy and energy entropy: Discrete wavelet transform also decomposes current signals of the fault phase into 3 wavelet scales, in which wavelet energy and energy entropy are calculated for characterization.

$$E_j = \sum |C_j|^2, S_j = -\frac{E_j}{\sum_j E_j} \log_2\left(\frac{E_j}{\sum_j E_j}\right) \quad (4)$$

where C_j denote the wavelet coefficient in scale j , $j = 1, 2, 3$.

- (6) Time domain factors: In order to reflect the waveform shape and shock characteristics of fault current, crest factor, form factor, skewness and kurtosis, denoted as t_1 – t_4 , are introduced to describe the fault-phase current.

$$t_1 = \frac{\max(i_n)}{\frac{1}{N_s} \sum_{n=1}^{N_s} |i_n|}, t_2 = \frac{\sqrt{\frac{1}{N_s} \sum_{n=1}^{N_s} (i_n)^2}}{\frac{1}{N_s} \sum_{n=1}^{N_s} |i_n|},$$

$$t_3 = \frac{\sum_{n=1}^{N_s} (i_n - \bar{i})^3}{SD^3 N_s}, t_4 = \frac{\sum_{n=1}^{N_s} (i_n - \bar{i})^4}{SD^4 N_s} \quad (5)$$

where SD is the standard deviation.

- (7) Approximation constant: According to [10], (6) is used to approximate the root mean square (rms) waveform of neutral current. Neutral voltage is also considered in this paper.

$$f(t, \delta) = 1 - e^{-\delta t} \quad (6)$$

where t denotes the time step and δ denotes the approximation constant.

- (8) Custom time constant of sequence current: Time content is defined as the time required to rise from the start to $1/e$ of the maximum amplitude. $1/e$ is substituted with a custom value $m \in [0.1, 0.2, 0.3, \dots, 0.9, 1]$ to reflect dynamic characteristics of current sequence components.
- (9) Fault inception phase angle (FIPA): With a uniform frequency this feature can be calculated as the time from the last zero crossing point to the fault triggering point.

Although high-resolution signals may contain enough transient information in theory, data integrity, sample quantity and sampling rate still restrict waveform characterization. For practical purpose, it is necessary to take contextual features into account. The increase of contextual features benefits from the development of monitoring and communication technologies, but most context data work independently and keep interconnected at present. Due to data restriction, this paper considers context data that are available in common. They are time stamp, geographical data, meteorological data, protection data and query information. After necessary discretization, these context data are then integrated as the pool of candidate contextual features. Pools of candidate features are shown in Table 1 and screening out irrelevant features is the focus in follow-up work.

Table 1. Pools of candidate features.

Type	Candidate features	Feature number
Waveform	Maximum sequence voltage/current	6
	Maximum change of three-phase signals and sequence components	6
	Sequence component values	24
	DC and harmonic content	6
	Wavelet energy and energy entropy	6
	Crest factor, form factor, skewness and kurtosis	4
	Approximation constants	2
	Custom time constant of sequence current	30
	FIPA	1
Context	Time stamp: season, day/night	2
	Location: landform, zone	2
	Meteorological data: weather, rainfall, cloud cover, temperature, humidity, wind scale, maximum wind speed	7
	Protection data: reclosing, fault type, fault duration, tripping time, reclosing time, breaker quenching time, number of triggering	7
	Others: voltage level, number of history faults	2

3. Classification model based on multi-view sparse feature selection

3.1. The proposed MVSFS method

To get over the shortcoming of feature concatenation, this paper proposes a multi-view feature selection method based on ε -dragging technology and sparse learning. We denote $\mathbf{X}_v = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n]^T \in \mathbb{R}^{n \times d_v}$ as the original feature matrix for view v , where n is the total number of fault instances and d_v is feature number in view v , $v = 1, 2$ in this paper. Then the label matrix is denoted as $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n]^T \in \{0, 1\}^{n \times c}$ with c categories of fault cause. Weight matrix $\mathbf{W}_v$ is used to evaluate view feature importance and denoted as $\mathbf{W}_v = [\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{d(v)}]^T \in \mathbb{R}^{d_v \times c}$.

Given the two-view fault dataset $(\mathbf{X}_1, \mathbf{X}_2, \mathbf{Y})$, the problem of feature selection in sparse learning can be transformed into determining the weight matrix for each view through an optimization problem and rank features accordingly. Higher ranked features work better in the classification. The optimization problem is formulated as

$$\min_{\mathbf{W}_1, \mathbf{W}_2} \sum_{v=1}^2 \|\mathbf{X}_v \mathbf{W}_v - \mathbf{Y}\|_F^2 + \alpha \|\mathbf{W}_1\|_{2,1} + \beta \|\mathbf{W}_2\|_{2,1} \quad (7)$$

$\sum_{v=1}^2 \|\mathbf{X}_v \mathbf{W}_v - \mathbf{Y}\|_F^2$ is the loss function and the others act as the penalty terms. $l_{2,1}$ -norm penalty can turn down the weights of unimportant features and assign large number numbers for important features. α and β are the nonnegative penalty parameters. Considering the nature of multi-view features, norm penalty and parameter together contribute to the hierarchical feature selection. $l_{2,1}$ -norm regulates weight matrixes at a lower level by row sparsity while α and β can adjust weight balance between two views at a higher level.

Furthermore, ε -dragging is performed in order to enlarge class distance for multiclass classification [12], which transforms the positive elements in $\mathbf{Y} \in \{0, 1\}^{n \times c}$ from 1 into $1 + \varepsilon_i$ and negative ones from 0 into $-\varepsilon_i$. ε s are variable elements in a nonnegative matrix noted as $\mathbf{M} \in \mathbb{R}^{n \times c}$. A constant matrix $\mathbf{B} \in \{-1, 1\}^{n \times c}$ represents dragging direction and is defined as

$$B_{ij} = \begin{cases} +1, & Y_{ij} = 1 \\ -1, & Y_{ij} = 0 \end{cases} \quad (8)$$

$\mathbf{B}$ and $\mathbf{M}$ are connected via the Hadamard product operator $\otimes$ and we obtain the new label matrix $\mathbf{Y}' = \mathbf{Y} + \mathbf{B} \otimes \mathbf{M}$. According to [13], $\|\mathbf{W}_v\|_{2,1}$ need to be relaxed by $Tr(\mathbf{W}_v^T \mathbf{C} \mathbf{W}_v)$ for further optimization. Finally the final objective function can be written as

$$\min_{\mathbf{W}_1, \mathbf{C}_1, \mathbf{W}_2, \mathbf{C}_2, \mathbf{M}} \sum_{v=1}^2 \|\mathbf{X}_v \mathbf{W}_v - \mathbf{Y} - \mathbf{B} \otimes \mathbf{M}\|_F^2 + \alpha Tr(\mathbf{W}_1^T \mathbf{C}_1 \mathbf{W}_1) + \beta Tr(\mathbf{W}_2^T \mathbf{C}_2 \mathbf{W}_2) \quad (9)$$

where $\mathbf{C}$ is a diagonal matrix with elements $c_{ii} = 1/(2\|\mathbf{w}\|_2)$ and $\mathbf{w}$ is the row vector of $\mathbf{W}_v$.

Though more variables are introduced for optimization, we get a convex function so the problem can be solved iteratively. We first consider the derivative *w.r.t* $\mathbf{W}_1$. Variables $\mathbf{M}$ and $\mathbf{C}_1$ are fixed in (10) after irrelevant items are removed.

$$\min_{\mathbf{W}_1} \|\mathbf{X}_1 \mathbf{W}_1 - \mathbf{Y} - \mathbf{B} \otimes \mathbf{M}\|_F^2 + \alpha Tr(\mathbf{W}_1^T \mathbf{C}_1 \mathbf{W}_1) \quad (10)$$

The derivative of (10) *w.r.t* $\mathbf{W}_1$ is given as

$$2\mathbf{X}_1^T (\mathbf{X}_1 \mathbf{W}_1 - \mathbf{Y} - \mathbf{B} \otimes \mathbf{M}) + 2\alpha \mathbf{C}_1 \mathbf{W}_1 \quad (11)$$

Let (11) equal to zero and solve the equation to update $\mathbf{W}_1$. $\mathbf{C}_1$ can be updated subsequently since it is directly derived from $\mathbf{W}_1$. $\mathbf{W}_2$ and $\mathbf{C}_2$ can be obtained following the same procedure. As for variable $\mathbf{M}$, we let $\mathbf{P} = \mathbf{X}_1 \mathbf{W}_1 - \mathbf{Y}$ and $\mathbf{Q} = \mathbf{X}_2 \mathbf{W}_2 - \mathbf{Y}$ after removing irrelevant items. Next we fix $\mathbf{P}$ and $\mathbf{Q}$, and decoupled the function (12) into $n \times c$ subfunctions that can be represented as (13).

$$\min_{\mathbf{M}} \|\mathbf{P} - \mathbf{B} \otimes \mathbf{M}\|_F^2 + \|\mathbf{Q} - \mathbf{B} \otimes \mathbf{M}\|_F^2 \quad (12)$$

$$\min_{M_{ij}} (P_{ij} - B_{ij} M_{ij})^2 + (Q_{ij} - B_{ij} M_{ij})^2 \quad (13)$$

Considering $B_{ij}^2 = 1$, we transform (13) into (14).

$$\min_{M_{ij}} (P_{ij} B_{ij} - M_{ij})^2 + (Q_{ij} B_{ij} - M_{ij})^2 = \min_{M_{ij}} P_{ij}^2 + Q_{ij}^2 + 2M_{ij} [M_{ij} - B_{ij} (P_{ij} + Q_{ij})] \quad (14)$$

Given $M_{ij} \geq 0$, M_{ij} can be calculated as

$$M_{ij} = \begin{cases} 0, & B_{ij}(P_{ij} + Q_{ij}) \leq 0 \\ \frac{1}{2}B_{ij}(P_{ij} + Q_{ij}), & B_{ij}(P_{ij} + Q_{ij}) > 0 \end{cases} \quad (15)$$

As the result, matrix $\mathbf{M}$ is updated as

$$\mathbf{M} = \text{maximum} \left(\frac{\mathbf{B} \otimes (\mathbf{P} + \mathbf{Q})}{2}, 0 \right) \quad (16)$$

After the iteration update rule for all variables is determined, an optimization algorithm is developed to solve the regression model and realize feature selection, as shown in Algorithm 1. We measure the importance of features for row vectors of $\mathbf{W}_1$ and $\mathbf{W}_2$, and rank them from high to low by l_2 -norm values. The top ranked features can be used for fault-cause classification with ML models.

Algorithm 1. The optimization algorithm for the MVSFS method

Input: $\mathbf{X}_1 \in \mathbb{R}^{n \times d_1}$; $\mathbf{X}_2 \in \mathbb{R}^{n \times d_2}$; $\mathbf{Y} \in \{0,1\}^{n \times c}$; α and β

Output: $\mathbf{W}_1 \in \mathbb{R}^{d_1 \times c}$; $\mathbf{W}_2 \in \mathbb{R}^{d_2 \times c}$

1: Calculate $\mathbf{B}$ from $\mathbf{Y}$ via (8)

2: Initialize $\mathbf{W}_1$, $\mathbf{W}_2$ and $\mathbf{M}$

3: Initialize $t = 0$

4: **Repeat**

5: Calculate $\mathbf{C}'_1$ from $\mathbf{W}'_1$

6: $\mathbf{W}'^{t+1}_1 = (\mathbf{X}_1^T \mathbf{X}_1 + \alpha \mathbf{C}'_1)^{-1} (\mathbf{X}_1^T \mathbf{Y} + \mathbf{X}_1^T \mathbf{B} \otimes \mathbf{M}')$

7: Calculate $\mathbf{C}'_2$ from $\mathbf{W}'_2$

8: $\mathbf{W}'^{t+1}_2 = (\mathbf{X}_2^T \mathbf{X}_2 + \alpha \mathbf{C}'_2)^{-1} (\mathbf{X}_2^T \mathbf{Y} + \mathbf{X}_2^T \mathbf{B} \otimes \mathbf{M}')$

9: $\mathbf{M}_{t+1} = \text{maximum} (0.5 * \mathbf{B} \otimes (\mathbf{P} + \mathbf{Q}), 0)$

10: $t = t + 1$

11: Calculate residue via (7) with $\mathbf{Y}'$

12: **Until** convergence or maximum iteration number

4. Case study

4.1. Experiment settings

To evaluate the proposed method, 276 line fault samples were collected from the AC network above 110 kV of a certain city in Guangdong Province, China. Voltage and current signals were recorded by digital fault recorders (DFRs) and stored in offline files of common format for transient data exchange (COMTRADE). DFRs recorded fault signals with 5 kHz sampling frequency. Context information was collected through the operation and maintenance system. Besides data collection for two views, inspection reports of line patrol were attached to these fault samples and labeled the causes. In terms of root causes, the dataset consisted of 154 lighting, 75 external force, 28 floating object contact and 19 others. Waveform and contextual features in Table 1 were extracted from this dataset to form the multi-view feature pools, followed by z-score standardization. After that, the subsequent fault identification steps included feature selection and classification.

To verify the superiority of MVSFS, we applied different feature selection method for both single view and multi-view learning. In single view learning, two view features are input independently for selection and classification steps. The performance of single view learning for experimental comparison was represented by the performance of the best single view (BSV). Multi-view learning conducted feature selection using both waveform and contextual features and made a classification with the optimized feature set. Two view features were concatenated as one input vector for several typical feature selection methods that are designed for single view learning. They were analysis of variance (ANOVA), joint mutual information (JMI), joint mutual information maximization (JMIM),

HSIC Lasso [14], ReliefF and recursive feature elimination (RFE). Besides, KCCA [15] and DCA [16] were introduced on behalf of multi-view learning and feature mapping.

As for classification, we adopted random forest (RF) as the classifier and developed a RF model with scikit-learn default setting plus sample weighting, carrying comparison experiments with features selected in the previous step. 70% samples of the dataset were adopted for training on every trial and the rest formed the test set. Hyperparameters α and β are tuned within $\{10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3\}$ via grid search on training sets. Classification accuracies of test sets were averaged over 10 trials as the performance results.

4.2. Results and analysis

The classification performance with increasing selected feature numbers is presented in Fig. 1. The experiments without feature selection or mapping selected all available features for classification, denoted as RAW in Fig. 1. Since almost all comparison methods are designed for single view data, their performance with two view feature is apparently worse than that in single view learning, which demonstrates our hypothesis and one of conclusions in [8]. Also, typical multi-view mapping method KCCA performs even worse. By introducing multi-view sparse representation, our feature selection method improves classification performance and achieves higher accuracies compared with other methods. As seen from Fig. 1(a), MVSFS continuously outperforms single view methods within 10~18 selected fault features and achieves the global optimum. When features keep increasing, the accuracies fall so other methods surpass the MVFS method with relatively converged accuracies. According to Fig. 1(b), only ANOVA performance under distributed certain feature number can surpass MVSFS whose overall performance is apparently better than all competitors using two view features.

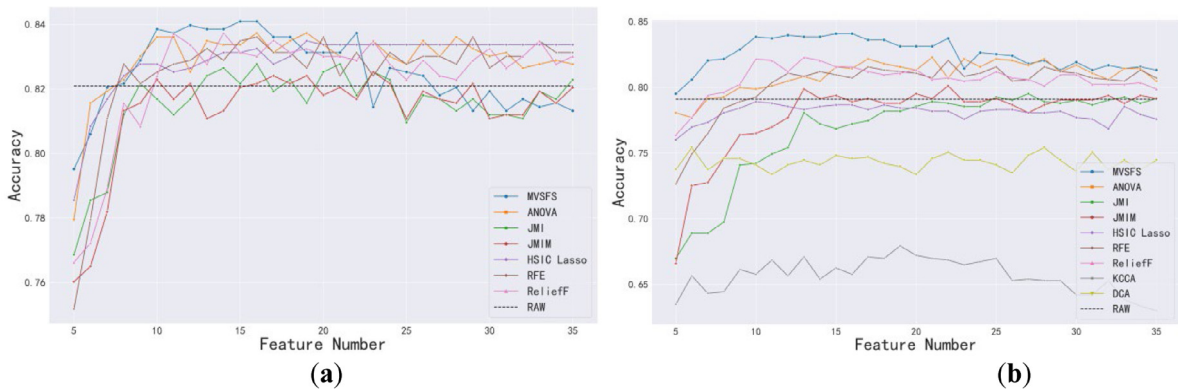


Fig. 1. Classification comparison between MVSFS and other feature algorithms with: (a) best single view; (b) two views.

Through the above experiments we notice context view is the best view in this dataset. Furthermore, we performed sensitivity study to explore the parameter effects on two views. Hyperparameters α and β were changed and tested ranging in $\{10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3\}$ and the highest accuracies for each trail are shown in Fig. 2. The parameter α corresponds to context view regularization and β corresponds to waveform view regularization. As seen from Fig. 2(a), the change of β made a different to average performance and higher β led to better performance while α made little different. One of explanations is that the less informative view need enhanced sparsity to weaken irrelevant features. By contrast, the context view carrying major discriminative features tends to cause overfitting with excessive sparsity when $\alpha = 100$.

In order to explore the effect of the classifiers, we replaced RF with other ML models, namely DT, k-nearest neighbor (KNN), support vector machine (SVM), extra tree (ET), dynamic ensemble selection (KNORAU) and stacking ensemble (StackedClassifier). The last two ensemble models were developed using machine learning library DESlib while others using scikit-learn. Four outstanding feature selection methods in two-view feature selection were employed and their best classification performance on different classifiers is presented in Table 2.

From the perspective of accuracy, ensemble models achieve better performance than base learners and MVSFS outperform others except on DT and KNN, which indicates MVSFS has better applicability for high-generalization learners. From the perspective of feature number, MVSFS and ReliefF are more stable than ANOVA and RFE.

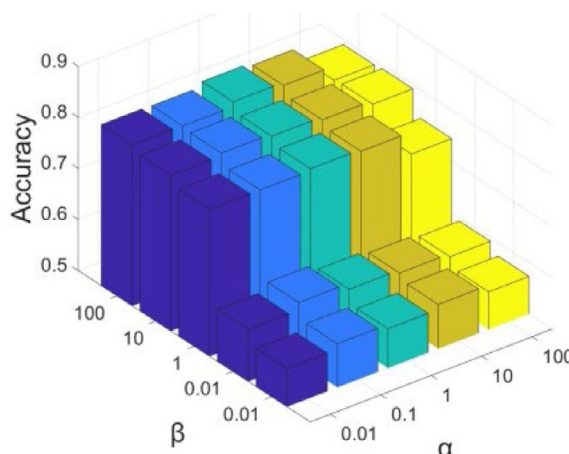


Fig. 2. Performance variation of MVSFS with different values of the parameters α and β .

Table 2. Best classification performance on different ML classifiers with two-view features.

Classifier	Feature selection							
	MVSFS		ANOVA		ReliefF		RFE	
	Accuracy	Feature number	Accuracy	Feature number	Accuracy	Feature number	Accuracy	Feature number
DT	0.772	11	0.760	35	0.778	7	0.734	16
KNN	0.783	16	0.795	7	0.775	6	0.752	14
SVM	0.806	22	0.765	6	0.767	14	0.726	34
RF	0.841	15	0.822	28	0.823	13	0.820	22
ET	0.852	17	0.829	25	0.822	15	0.819	28
KNORAU	0.851	16	0.826	28	0.817	10	0.807	18
StackedClassifier	0.860	16	0.831	25	0.815	15	0.817	35

ReliefF always takes less features for best performance but its accuracies are limited. The number range of MVSFS is smaller than those of ANOVA and RFE, and on high-accuracy classifiers, best feature numbers are even closer, ranging from 15–17. A conclusion can be drawn that MVSFS can achieve superior performance with high-generalization classifiers.

To sum up, a new method for the combined use of contextual and waveform features has been proven to improve the performance for fault-cause classification. Compared with single view methods, the MVSFS method can continuously achieve better accuracy with fewer fault features; it also reveals overall superiority against comparison methods using two view data and shows better applicability for high-generalization learners. Furthermore, context view tends to carry more discriminative features because many hand-crafted waveform features are irrelevant or redundant. More fault signature studies are expected to improve hand-craft features and refine the feature pool.

5. Conclusion

In order to combine contextual and waveform data for better fault-cause classification, a multi-view sparse feature selection method is proposed to tackle the fusion challenge. In our sparse regression model, $l_{2,1}$ -norm penalty is designed to select features across views with joint sparsity and an ε -dragging technique is used to enlarge label distance. The model can be solved by an iteration optimization algorithm. Experiments on real-life fault data demonstrate that contextual and waveform features are able to achieve higher accuracy than single view learning only through appropriate fusion, and the proposed MVSFS that overcomes the drawback of conventional feature selection methods can improve the classification performance with multi-view data.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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